

Dustline (Desert / Military) Map Design

Executive Summary

Dustline is a wide-open desert military battlefield built for high-mobility warfare, long-range engagements, and rapid reconnaissance. The terrain features rolling dunes, hardened military outposts, abandoned airstrips, dry canyons, and exposed open flats. The map emphasizes Obsidian Protocol's autonomous maneuvering, sensor-driven recon, comms-relay warfare, and logistics stretched across open terrain. In AR (Real Light/VR), the map anchors at ~4-6 m viewing distance, covering a 10x10 m physical area representing ~1 km² of desert military territory. Key features include mobile warfare corridors, elevated ridges for sensors, open flats for vehicle combat, and critical military infrastructure.

Design Goals & Gameplay

Mobile Desert Warfare

Dustline prioritizes fast vehicle movement, long-range shooting lanes, and rapid repositioning. Open terrain favors armor, recon vehicles, and drone support.

Autonomy & Maneuver Logic

Autonomous AI selects optimal movement routes across dunes, avoids exposed flats during enemy artillery fire, and uses ridges for recon vantage.

Sensor & Information Warfare

Dust storms, heat shimmer, and open terrain distort sensors. Relay towers on ridges restore clarity. Destroying relays creates massive fog-of-war zones.

Logistics & Resources

Supply convoys avoid routes exposed roads vulnerable to ambush. Controlling the Airstrip and Military Depot provides major resupply advantages.

Deployment & Progress

A hard deployment budget (~10,000 points) forces players to choose between heavy armor or fast recon units. Blue deploys from the southwest dune base. Red deploys from the northeast military ridge.

Spatial Constraints (Nreal AR)

Device Capabilities

Target Lightbox: "5.5" diagonal FOV, 1800x1080 per-eye resolution. Optimal viewing distance ~4-6 m.

Playable Area & Scaling

10x10 m physical area is 1 km² desert military zone (1,000 units). Designed for tabletop AR.

Tracking & Occlusion

100+ m physical area is 1 km² desert military zone. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

Key Locations

Blue HQ (Dune Base) — (0, 0). 120x120 m reinforced dune-side base with armored vehicles.

Red HQ (Military Ridge) — (1000, 1000). 120x120 m elevated ridge with radar mast and hardened bunkers.

Central Flat — (500, 500). 300x300 m Open desert floor. Ideal for mobile vehicle combat.

Military Depot — (120, 800). 80x80 m Hardly crater. Fuel tanks, armored trucks.

Abandoned Airstrip — (800, 300). 150x150 m Runway. Bombers, radar dishes, drone pads.

Dry Canyon Base — (0, 400). 4x 400 300 m Natural gully. Better cover with fortified cover.

Ridge A — (600, 150). ~250 m Long narrow sensor vantage point.

Ridge B — (150, 650). ~150 m Gully-side sensor tower and recon hub.

Mobile Warfare Corridor — (300, 0) to (400, 300) Fast movement path for vehicles.

Neutral Forward Outpost — (450, 850) Open neutral zone.

Chokepoints & Sightlines

Drv Canyon Route

Provides point cover. Key flanking path.

Airstrip Hanger Zone

Close range combat inside hangars; long range outside.

Central Flats

Extends long range sightlines; armor and artillery dominates.

Sightlines

Ridge A provides long range sensor coverage (~300 m). Dust storms create fog-of-war pockets.

Deployment Zones

Blue Deployment: HQ dune base + rear rally point.

Red Deployment: HQ military ridge + rear rally point.

Neutral Forward Zone: Outpost at (500, 850) for drone drops.

Assets & Environment

Terrain

Rolling dunes, rocky ridges, dry canyons, open flats.

Vegetation

Scattered shrubs, dry grass patches.

Buildings & Props

Military: Hangars, bunkers, radar dishes, watchtowers. Industrial: Fuel tanks, generators, pipelines. Logistics: Trucks, crates, fuel drums, recon drones.

Effects

Dust storms, heat shimmer, smoke plumes, wind gusts.

Audio

Wind, distant machinery, radio chatter, sand movement.

LOD & Budget

Asset List (Count)

LOD/Budget

Assets

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